HK IMO Training 2023-2024
Problem Set 15 Suggested Solutions

Problem 57 (Chile MO 2021 Problem 3). Find all polynomials $P(x)$ with real coefficients satisfying

$$4P(x)^2 + 4P(x) = 4P(x^2) + 1.$$

Solution. Answer: $P(x) = x^k - \frac{1}{2}$ for some $k \geqslant 0$ or $P(x) = -\frac{1}{2}$.

Note that

$$4\left(P(x) + \frac{1}{2}\right)^2 = 4P(x)^2 + 4P(x) + 1 = 4P(x^2) + 2 = 4\left(P(x^2) + \frac{1}{2}\right).$$

Let $Q(x) = P(x) + \frac{1}{2}$. It remains to solve $Q(x)^2 = Q(x^2)$.

- Clearly, $Q(x) = 0$ is one solution. This corresponds to $P(x) = -\frac{1}{2}$.
- If $Q(x) = ax^k$ for some $a \neq 0$ and $k \geqslant 0$, the equation becomes $a^2x^{2k} = ax^{2k}$. This holds iff $a^2 = a$, or $a = 1$. Therefore, the solutions in this case are $Q(x) = x^k$, which correspond to $P(x) = x^k - \frac{1}{2}$ for $k \geqslant 0$.
- If $Q(x) = ax^k + bx^\ell + R(x)$ for some $a, b \neq 0$ and $k > \ell > \deg R$, the equation becomes

  $$a^2x^{2k} + 2abx^{k+\ell} + \cdots = ax^{2k} + bx^{2\ell} + \cdots,$$

  where $\cdots$ includes terms of smaller degree. Note that the coefficient of $x^{k+\ell}$ on the left is $2ab \neq 0$, while that on the right is 0 as $2k > k + \ell > 2\ell$. This is a contradiction. So there is no solution in this case.

Summarizing these cases, the only solutions are those stated at the beginning. $\square$

Problem 58 (Cono Sur Olympiad 2019 Problem 5). Let $n \geqslant 3$ be an integer. In how many ways can we put down 0 or 1 in each cell of an $n \times n$ table such that the sum of all six numbers in every 2×3 or 3×2 rectangle is even?

Solution. Answer: 32 ways.

We use a_{ij} to denote the number in the (i, j)th cell. All numbers in the cells are taken modulo 2. The requirements are

$$\begin{aligned} P(i,j): &\quad a_{ij} + a_{i(j+1)} + a_{i(j+2)} + a_{(i+1)j} + a_{(i+1)(j+1)} + a_{(i+1)(j+2)} = 0 \text{ where } i \leqslant n-1, j \leqslant n-2, \\ Q(i,j): &\quad a_{ij} + a_{(i+1)j} + a_{(i+2)j} + a_{i(j+1)} + a_{(i+1)(j+1)} + a_{(i+2)(j+1)} = 0 \text{ where } i \leqslant n-2, j \leqslant n-1. \end{aligned}$$

We are going to show that all entries are uniquely determined by the values of $a_{11}, a_{12}, a_{21}, a_{22}, a_{33}$, and these five entries are independent, making a total of $2^5 = 32$ possibilities. Firstly, $P(1,1) + P(2,1) + Q(1,1)$ gives $a_{13} + a_{21} + a_{22} + a_{33} = 0$. This implies

$$a_{13} = a_{21} + a_{22} + a_{33}. \tag{1}$$

Similarly, from $P(1,1) + Q(1,1) + Q(1,2)$, we obtain

$$a_{31} = a_{12} + a_{22} + a_{33}. \tag{2}$$

From $P(1,1)$ and (1), we have

$$a_{23} = a_{11} + a_{12} + a_{33}. \tag{3}$$

From $Q(1,1)$ and (2), we have

$$a_{32} = a_{11} + a_{21} + a_{33}. \tag{4}$$

Conversely, if $a_{13}, a_{31}, a_{23}, a_{32}$ are defined using (1) to (4), then it is easy to check that $P(1,1), P(2,1), Q(1,1), Q(1,2)$ are satisfied. This implicitly solves the problem when $n = 3$.

We now assume $n \geqslant 4$. We prove that $a_{ij} = a_{i(j+3)}$ for all $i \leqslant n$ and $j \leqslant n-3$. Firstly, the sum $P(i,j)+P(i,j+1)$ gives $a_{ij}+a_{i(j+3)}+a_{(i+1)j}+a_{(i+1)(j+3)}=0$. This implies

$$a_{ij}+a_{i(j+3)} = a_{(i+1)j}+a_{(i+1)(j+3)}$$

for all $i \leqslant n-1$ and $j \leqslant n-3$. It follows that

$$a_{ij}+a_{i(j+3)} = a_{1j}+a_{1(j+3)} \tag{5}$$

for all $i \leqslant n$ and $j \leqslant n-3$. Secondly, the sum $Q(i,j)+Q(i,j+1)+Q(i,j+2)$ gives $a_{ij}+a_{i(j+3)}+a_{(i+1)j}+a_{(i+1)(j+3)}+a_{(i+2)j}+a_{(i+2)(j+3)}=0$. Together with (5), we obtain $a_{1j}+a_{1(j+3)}=0$, and hence

$$a_{ij}+a_{i(j+3)}=0$$

for all $i \leqslant n$ and $j \leqslant n-3$. This means exactly $a_{ij}=a_{i(j+3)}$. By symmetry, we also have $a_{ij}=a_{(i+3)j}$ for all $i \leqslant n-3$ and $j \leqslant n$. Therefore, the values of all a_{ij} can be determined by $a_{11}, a_{12}, a_{21}, a_{22}, a_{33}$.

Conversely, if $a_{ij}=a_{i(j+3)}$ and $a_{ij}=a_{(i+3)j}$, we show that all conditions are satisfied. Since

$$\begin{aligned}
&\left(a_{ij}+a_{i(j+1)}+a_{i(j+2)}+a_{(i+1)j}+a_{(i+1)(j+1)}+a_{(i+1)(j+2)}\right)\\
&\ +\left(a_{(i+1)j}+a_{(i+1)(j+1)}+a_{(i+1)(j+2)}+a_{(i+2)j}+a_{(i+2)(j+1)}+a_{(i+2)(j+2)}\right)\\
&\ +\left(a_{(i+2)j}+a_{(i+2)(j+1)}+a_{(i+2)(j+2)}+a_{(i+3)j}+a_{(i+3)(j+1)}+a_{(i+3)(j+2)}\right)\\
&=\left(a_{ij}+a_{(i+3)j}\right)+\left(a_{i(j+1)}+a_{(i+3)(j+1)}\right)+\left(a_{i(j+2)}+a_{(i+3)(j+2)}\right)\\
&=0,
\end{aligned}$$

the condition $P(i+2,j)$ holds iff $P(i,j)$ and $P(i+1,j)$ hold. Similarly, $Q(i,j+2)$ holds iff $Q(i,j)$ and $Q(i,j+1)$ hold. Therefore, it suffices to show that $P(1,1), P(2,1), Q(1,1), Q(1,2)$ hold. But we have shown that these hold as long as (1) to (4) are satisfied. This completes the proof. $\square$

Problem 59 (Cono Sur Olympiad 2022 Problem 3)**.** Let k be a positive integer. Prove that there exists a positive integer n such that the decimal representation of each of the numbers $n, n^2, \ldots, n^k$ contains the consecutive digits 2024.

Solution. We prove this by induction on k. When $k=1$, we can take $n=20241$. Suppose there exists a positive integer m with $(m,10)=1$ such that each of the numbers $m, m^2, \ldots, m^{k-1}$ contains 2024. We shall prove that there exists a positive integer n with $(n,10)=1$ such that each of the numbers $n, n^2, \ldots, n^k$ contains 2024.

Let $k=2^a5^bc$ where $(c,10)=1$. Consider

$$n = 10^M 2^b 5^a d + m$$

where $d \equiv 20241(cm^{k-1})^{-1} \pmod{10^5}$, and M is a sufficiently large positive integer. Note that $(n,10)=(m,10)=1$. For $j \leqslant k-1$, we have

$$n^j = (10^M 2^b 5^a d + m)^j = 10^M A + m^j$$

for some positive integer A. Since M is sufficiently large, n^j contains the number m^j in its tail part, which includes 2024 by the inductive hypothesis. Also, we have

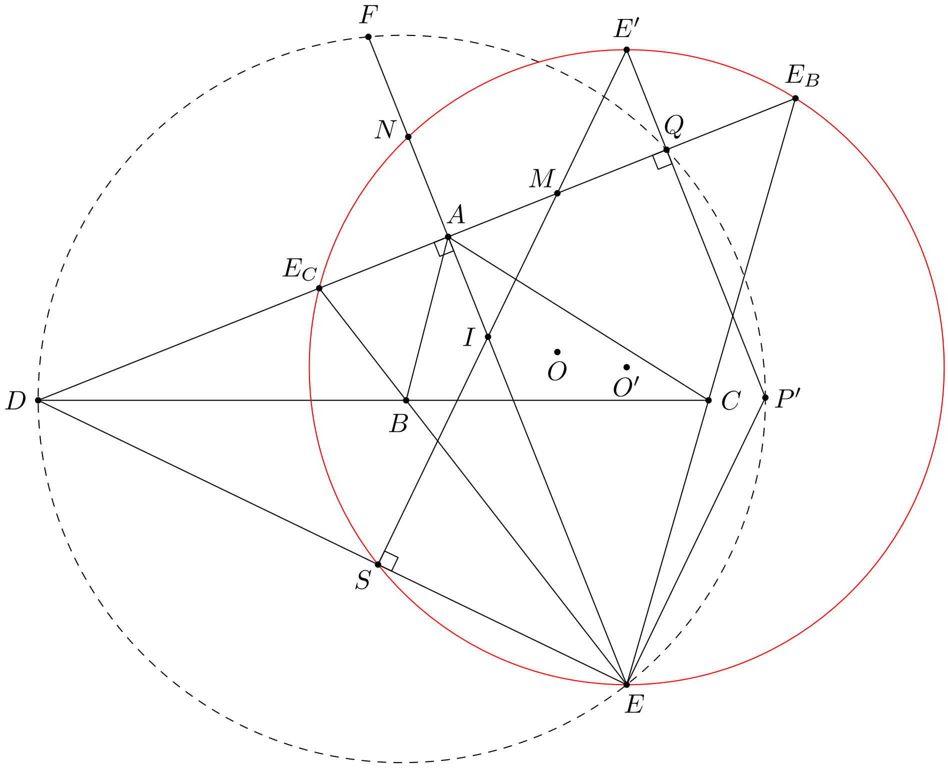
$$n^k = (10^M 2^b 5^a d + m)^k = 10^{2M} B + k \cdot 10^M 2^b 5^a d m^{k-1} + m^k$$

for some positive integer B. So n^k contains the number $k \cdot 2^b 5^a d m^{k-1} = 10^{a+b} c d m^{k-1}$. As

$$cdm^{k-1} \equiv 20241 \pmod{10^5}$$

by construction, n^k contains the number 2024. This proves the inductive step, and so the result follows readily. $\square$

Problem 60 (Iran MO 2020 3rd Round G4). In $\triangle ABC$, let O, I and E be the circumcentre, the incentre and the A-excentre respectively. Let D be the intersection point of the line BC and the external angle bisector of $\angle BAC$. Let F be a point on the extension of IA beyond A such that $AF = 2IA$. Define P to be the point on the circumcircle of $\triangle DEF$ such that DP is a diameter. Prove that $OP = 3OI$.



Solution. Let E_B and E_C be the B-excentre and the C-excentre of $\triangle ABC$ respectively, which lie on the line AD. We first focus on $\triangle EE_BE_C$. Note that I is its orthocentre and A, B, C are the feet of altitudes. Let O' be its circumcentre. Also, define the following points.

- E' and P' are the reflections of E and I about O' respectively
- M is the midpoint of E_BE_C
- N is the reflection of I about E_BE_C (which lies on (EE_BE_C), and N is the midpoint of AF)
- Q is the intersection point of E_BE_C and $E'P'$

It is well-known that E', M, I are collinear, and $E'MI$ meets DE at a point S on (EE_BE_C).

We first prove that D, E, P', Q are concyclic. As O' is the common midpoint of IP' and EE', $IEP'E'$ is a parallelogram. Therefore, we have

$$\angle QP'E = \angle E'P'E = \angle EIE' = \angle AIS = 180^\circ - \angle EDQ.$$

This implies D, E, P', Q are concyclic.

Secondly, we prove that F, E, P', Q are concyclic. Note that $FE /\!/ QP'$ since $IEP'E'$ is a parallelogram. So it suffices to show that $FEP'Q$ is an isosceles trapezoid. Indeed, as $FN = NA = AI$ and $NAQE'$ is a rectangle (well-known), $FIQE'$ is an isosceles trapezoid, and hence $FQ = IE' = EP'$. This implies $FEP'Q$ is an isosceles trapezoid, and so it is cyclic.

Combining these, we find that D, E, P', Q, F are concyclic. As $\angle DQP' = 90^\circ$, DP' is a diameter of the circle passing through these points. Thus, we have proved that $P' = P$. Now, note that O is the nine-point centre of $\triangle EE_BE_C$. Therefore, O is the midpoint of IO'. As O' is the midpoint of IP, it is obvious that $OP = 3OI$. $\square$